

```
print("INITIAL STATE:")
display(initial)

print("GOAL STATE:")
display(goal)

solution = dfs(initial, goal)

if solution:

    print("\nDFS SOLUTION FOUND")
    print("Number of moves:", len(solution) - 1)

    print("\nSolution Path:")

    for step, state in enumerate(solution):

        print("\nStep", step)
        display(state)

else:
    print("No solution found.")
```

INITIAL STATE:

```
-----
| 1 | 2 | 3 |
| 4 | 0 | 6 |
| 7 | 5 | 8 |
-----
```

GOAL STATE:

```
-----
| 1 | 2 | 3 |
| 4 | 5 | 6 |
| 7 | 8 | 0 |
-----
```

DFS SOLUTION FOUND
Number of moves: 434

Solution Path:



```
print("INITIAL STATE:")
display(initial)

print("GOAL STATE:")
display(goal)

solution = dfs(initial, goal)

if solution:

    print("\nDFS SOLUTION FOUND")
    print("Number of moves:", len(solution) - 1)

    print("\nSolution Path:")

    for step, state in enumerate(solution):

        print("\nStep", step)
        display(state)

else:
    print("No solution found.")
```

```
| 1 | 2 | 3 |
| 4 | 5 | 6 |
| 0 | 7 | 8 |
-----
```

Step 433

```
| 1 | 2 | 3 |
| 4 | 5 | 6 |
| 7 | 0 | 8 |
-----
```

Step 434

```
| 1 | 2 | 3 |
| 4 | 5 | 6 |
| 7 | 8 | 0 |
-----
```

Home

8-Puzzle DFS

8-Puzzle I-DFS

1BM24CS173-[AI-LAB]

Vacuum Cleaner

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Jupyter 8-Puzzle I-DFS Last Checkpoint: 1 minute ago

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Trusted

JupyterLab Python 3 (ipykernel)

```
print(step, step)
display(state)
```

INITIAL STATE:

	1		2		3	
	4		0		6	
	7		5		8	

GOAL STATE:

	1		2		3	
	4		5		6	
	7		8		0	

Searching at depth: 0
Searching at depth: 1
Searching at depth: 2

IDS SOLUTION FOUND
Depth: 2
Number of moves: 2

Solution Path:

Step 0

	1		2		3	
	4		0		6	
	7		5		8	

Step 1

	1		2		3	
	4		5		6	
	7		0		8	

Step 2

	1		2		3	
	4		5		6	
	7		8		0	